

FL-06 – Design Your Personal Agent

Agent Design Document | General AI Fluency Track

STUDENT	Leon Infant B	AGENT NAME	FlyRank Research Scout
ASSIGNMENT CODE	FL-06	WEEK	5
ESTIMATED BUILD TIME	10 Hours	SELECTED PLATFORM	Python + Google Colab + VS Code

1 Job to Be Done

The purpose of the FlyRank Research Scout is to analyze SEO data and transform raw information into useful business insights. The agent helps users perform the following tasks:

- Analyze Google Search Console (GSC) data
- Analyze Google Analytics 4 (GA4) data
- Detect content opportunities
- Perform semantic clustering
- Classify search intent
- Generate reports and recommendations

The objective is to automate repetitive tasks and help users focus on decision-making instead of manual analysis.

2 User Profile

Primary User

Leon Infant B

Usage Frequency

- Daily during the hackathon
- Two to five sessions per day
- Weekly performance reviews

Goals

- Identify new opportunities for content growth

- Improve search performance
- Learn AI engineering concepts
- Build an intelligent workflow

3 Scope of the Project

This project is intentionally designed to remain small enough to complete within approximately ten hours.

INCLUDED	EXCLUDED
Data cleaning	Production deployment
Semantic clustering	Database administration
Intent classification	Continuous monitoring
Opportunity scoring	Automatic content publication
Insight generation	
Report generation	

4 Tools and Data Sources

Tools

Tool	Purpose	Access Method
Python	Data processing	Local installation
Pandas	Data manipulation	Open source
NumPy	Numerical calculations	Open source
Scikit-learn	Machine learning	Open source
Sentence Transformers	Embedding generation	Hugging Face
Google Colab	Development environment	Free version
GitHub	Version control	Existing account
OpenAI API	Intent classification	API access

Data Sources

Data Source	Purpose
Google Search Console	Search analysis
Google Analytics 4	User engagement analysis
Website content	Intent analysis
Historical records	Trend analysis

5 Agent Workflow



6 Draft Instructions

You are the FlyRank Research Scout. Your job is to analyze search data and generate actionable insights.

Follow these rules:

- Think step by step
- Explain your reasoning clearly
- Identify missing information

- Use available tools whenever necessary
- Prioritize business value
- Support every conclusion with evidence
- Highlight unusual patterns

ALWAYS DO	NEVER DO
Provide confidence scores	Fabricate information
Explain recommendations	Modify original data
Identify potential risks	Share confidential information
Produce concise reports	Delete files automatically

7 Evaluation Cases

Case 1

Input: A page receives a large number of impressions but very few clicks.

Expected Output: The agent should recommend title optimization and meta-description improvements.

Case 2

Input: Multiple pages compete for identical search terms.

Expected Output: The agent should identify content cannibalization.

Case 3

Input: Search queries include: "Magnesium taurate versus glycinate," "Alternative to Epsom salt," and "Best magnesium supplement."

Expected Output: The agent should classify comparison intent correctly.

Case 4

Input: A page receives significant traffic but shows low engagement.

Expected Output: The agent should identify an intent mismatch.

Case 5

Input: The agent detects an emerging search trend.

Expected Output: The agent should recommend the creation of new content.

8 Risks and Guardrails

Risk	Guardrail
Incorrect classifications	Use confidence scores
Missing information	Display warnings
Hallucinations	Require supporting evidence
Confidential data leakage	Restrict external access

Risk	Guardrail
Destructive operations	Require confirmation

Confirmation Requirements

The agent must confirm:

- Uploaded files
- Dataset structure
- Missing values
- Parameters used in scoring

Restricted Actions

The agent must never:

- Delete data automatically
- Share confidential information
- Modify original datasets
- Invent statistics

9 Platform Selection

Selected Platform

Python Script + Google Colab + VS Code

Reasons

- Completely free to use
- Flexible and scalable
- Compatible with machine-learning libraries
- Suitable for rapid development
- Easy integration with external tools

Alternative Platform: Custom GPT

ADVANTAGES	DISADVANTAGES
Simple interface	Requires a subscription
Easy setup	Limited flexibility

10 Success Criteria

The project will be considered successful if the agent can:

- Process uploaded datasets
- Detect search patterns
- Identify opportunities
- Generate recommendations
- Produce reports automatically

Conclusion

The FlyRank Research Scout will serve as an intelligent assistant capable of transforming search data into valuable business insights. By combining machine learning, semantic analysis, and AI-powered reasoning, the agent will reduce manual work and help identify high-impact opportunities more efficiently.